

11 Layout

11.1 Layout Guidelines

It is strongly recommended for best measurement performance to keep high current signals from interfering with the measurement system inputs and ground.

A second key recommendation is to ensure that the BQ769x0 input filtering capacitors and power capacitors are connected to a common ground with as little parasitic resistance between the connections as possible.

11.2 Layout Example

Figure 11-1 shows a guideline of how to place key components compared to respective ground zones, based on the BQ76920, BQ76930, and BQ76940 EVMs.

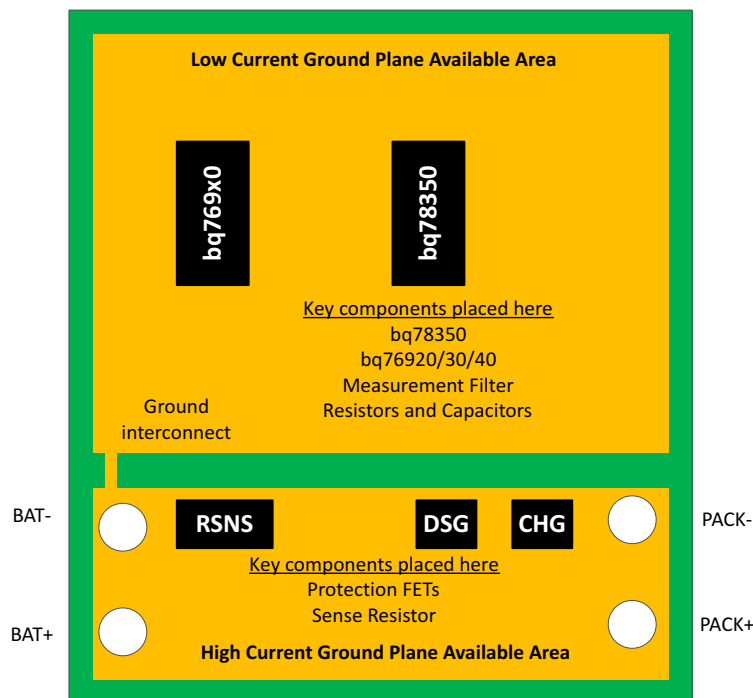


Figure 11-1. System Component Placement Layout vs. Ground Zone Guide

CAUTION

Care should be taken when placing key power pin capacitors to minimize PCB trace impedances. These impedances could result in device resets or other unexpected operations when the device is at peak power consumption.

Although not shown in the diagrams, this caution also applies to the resistor and capacitor network surrounding the current sense resistor.